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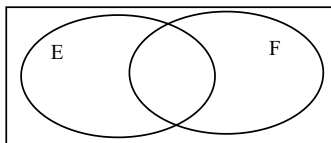
Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 7

Suppose $P(E) = 0.28$, $P(F) = 0.43$, and $P(E \cap F) = 0.17$. Use a Venn diagram to find the following.

- a. $P(E \cup F)$ b. $P(E' \cap F)$
 c. $P(E \cap F')$ d. $P(E' \cup F')$

Use the Venn diagram showing sets E and F.



Determine the probability for each region of the diagram. First find the probability for the region of E that does not overlap F.

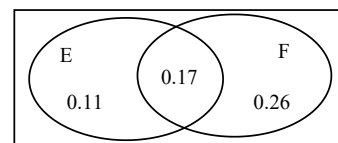
$$0.28 - 0.17 = 0.11$$

Now find the probability for the region of F that does not overlap E.

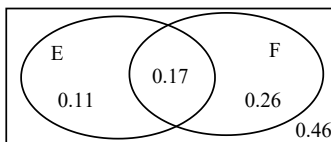
$$0.43 - 0.17 = 0.26$$

From the Venn diagram, the labeled regions have a total probability of $0.11 + 0.17 + 0.26 = 0.54$. Since the entire region of the Venn diagram must have probability 1, find the probability of the region outside E and F.

$$1 - 0.54 = 0.46$$



The completed Venn diagram is shown to the right.



a. Find $P(E \cup F)$. Substitute the given values into $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ and evaluate.

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$P(E \cup F) = 0.28 + 0.43 - 0.17$$

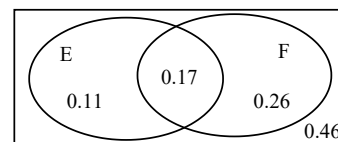
Substitute.

$$P(E \cup F) = 0.54$$

Simplify.

Thus, $P(E \cup F) = 0.54$.

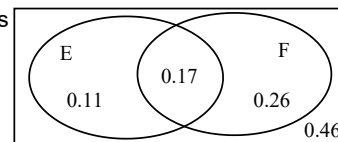
b. Find $P(E' \cap F)$. Look at E' , which includes the regions labeled 0.26 and 0.46. Then look at F, which includes the region labeled 0.26 and 0.17.



Find $P(E' \cap F)$. $E' \cap F$ is the region they have in common.

$$P(E' \cap F) = 0.26$$

c. Find $P(E \cap F')$. Look at E, which includes the region labeled 0.11 and 0.17. Then look at F' , which includes the regions labeled 0.11 and 0.46.



Find $P(E \cap F')$. $E \cap F'$ is the region they have in common.

$$P(E \cap F') = 0.11$$

d. Find $P(E' \cup F')$. The region $E' \cup F'$ includes regions that are not in E or are not in F.

Find $P(E' \cup F')$.

$$P(E' \cup F') = 0.46 + 0.11 + 0.26$$

$$= 0.83$$

